Teaching a Neural Network the Glow of the Solar System

A geometrically constrained model of zodiacal light for NASA's SPHEREx mission

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A geometrically constrained neural network learns the zodiacal light from viewing geometry alone, reaching ~0.02% median error on months it never saw.

Background

- **SPHEREx:** NASA space telescope (launched March 2025) mapping the **entire sky in infrared** (0.75–5 μm)
- **Science:** how the universe began, how galaxies formed, where water and other ingredients for life are found

Zodiacal light

- Sunlight scattered by dust from comets and asteroids, plus the dust's heat glow (Fig. 1)
- Far brighter than the faint glow of distant galaxies SPHEREx aims to measure
- Changes with pointing and Earth's place in orbit → **must be modeled and removed**

The Kelsall model

- Standard dust model from NASA's COBE; **ZodiPy** evaluates its prediction of the zodiacal-light contribution to what SPHEREx observes

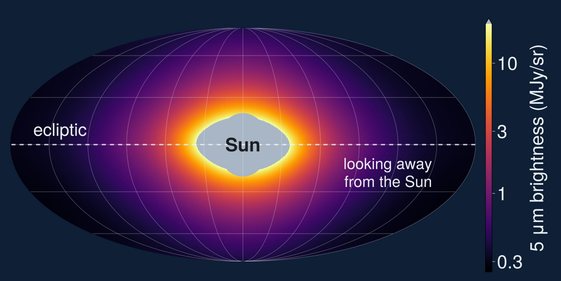


Figure 1. The whole sky at 5 μm from SPHEREx (ZodiPy): brightest toward the Sun and along the plane of the planets. Grey: too close to the Sun to observe.

The problem

- Zodiacal light must be removed from every SPHEREx image, but Kelsall's model uses a finite number of analytic components that don't fully describe the spatial and spectral structure in real data

Research objective

- A **geometrically constrained** neural network that learns zodiacal light from SPHEREx data and **outperforms the Kelsall model**. First step: ZodiPy's simulated sky

Methods

A geometrically constrained model

- No dates or sky coordinates: only **9 numbers describing the viewing geometry** (Fig. 2), dot products of the pointing vector n
- It must learn how brightness depends on geometry, as the physics does, not memorize when and where SPHEREx looked

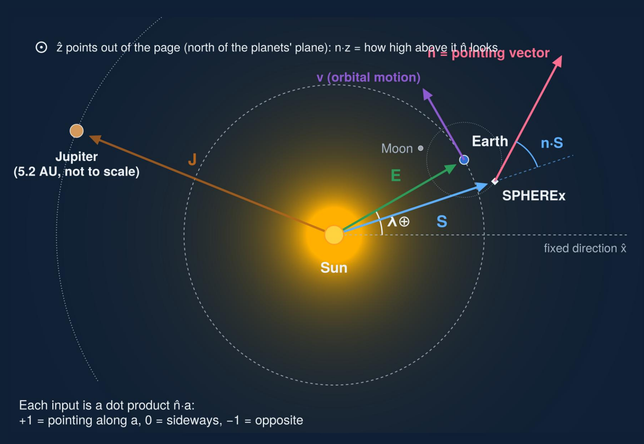


Figure 2. The geometry the network sees (top view, not to scale): n = SPHEREx's pointing vector; S, E = SPHEREx, Earth positions; v = orbital motion; λ⊕ = Earth's angle around its orbit. Jupiter and the Moon are tracked for real data: dust clouds leading and trailing Jupiter, and moonlight.

The 9 inputs

n·S, n·E	toward or away from the Sun (seen from SPHEREx and from Earth)
n·z	how far above or below the plane of the planets
n·vS, n·vE	ahead of or behind the orbital motion (SPHEREx, Earth)
n·x⊕, n·y⊕	Earth-following frame: x⊕ = Sun → Earth, y⊕ = Earth's motion
sin λ⊕, cos λ⊕	Earth's position in its orbit: the time of year

Training and testing

- 423,607 simulated pointings (140 days, 5 μm); five small networks (Flux.jl), averaged
- Tested on whole months and sky regions they never saw

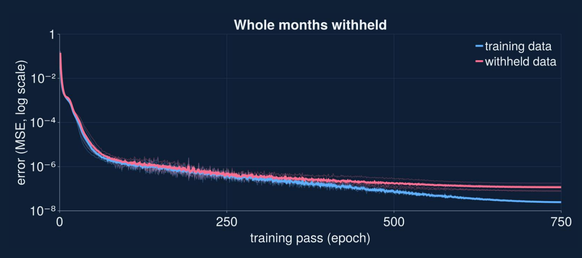


Figure 3. Error during training: 5 networks (thin), average (thick). The error on withheld months levels off and never rises: no overfitting.

Results

0.018% median error on months withheld from training

97.5% of pointings within 0.1% of ZodiPy

≤0.03% median error for every choice of withheld months

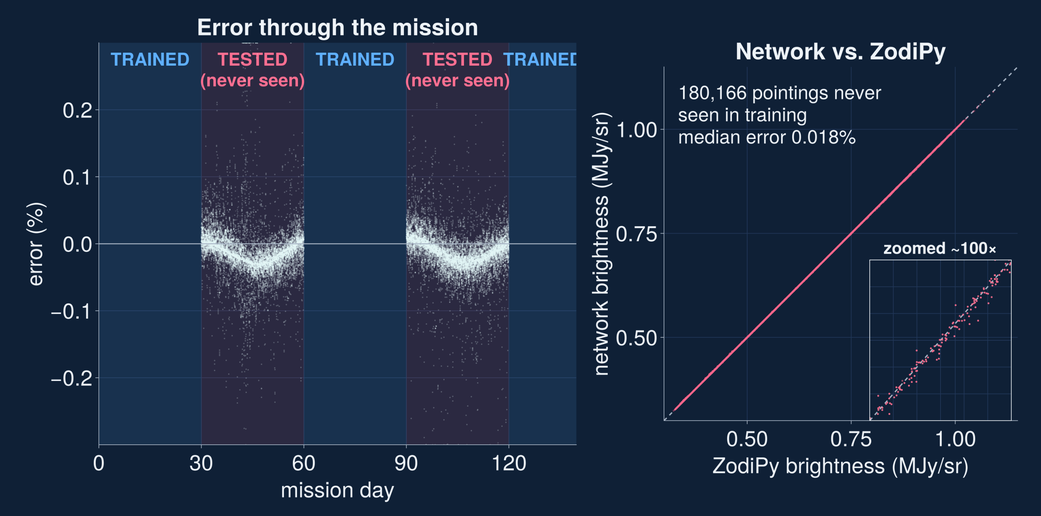


Figure 4. Network vs. ZodiPy on the two 30-day blocks never used in training (days 30–60, 90–120; 180,166 pointings). Left: error through the mission; right: every pointing falls on the 1:1 line.

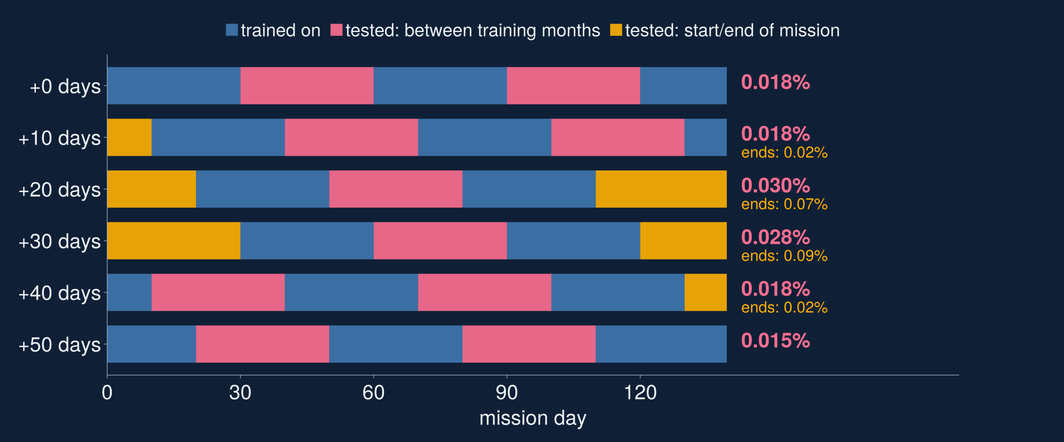


Figure 5. Six versions of the test, sliding the withheld months 10 days at a time: all reach 0.015–0.03% (red). Predicting past the start/end of the data (orange) is harder: fine for ~10 days, then errors grow.

- **Largest errors** along the plane of the planets, where the dust bands are; **smallest** toward the poles (Fig. 6)
- **More data:** with all 500 days the whole sky is tested and the plane-of-the-planets error drops (0.083% → 0.064%)

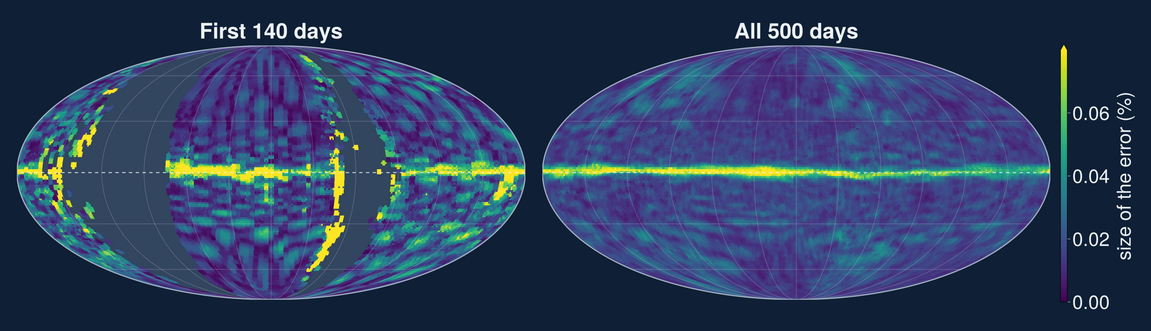


Figure 6. Error on pointings the network never saw, across the sky (3.5° patches, about SPHEREx's view). Left: the first 140 days. Right: all 500 days now available; with more than a year of data, Jupiter's direction (n·J) also helps the network tell the two years apart. Grey: not tested.

Next Steps

✓ DONE: Validate
Learn ZodiPy's simulated sky from geometry alone

IN PROGRESS: Refine
More data (500 days), longer training, Jupiter's direction

NEXT: Scale up
Move training to a computing cluster

NEXT: Real data
Train on real SPHEREx images; compare with Kelsall

All results so far are preliminary and use simulated data.

Acknowledgements

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References

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